

Amelia Chatterjee

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 97% (Mastered)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Python, PCB Debugging, MQTT, ARM Cortex, Embedded Linux, I2C/SPI/UART, FreeRTOS, C/C++

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Python service, reducing latency by 48%.
- Led a squad of 6 engineers delivering scalable microservices processing 44M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 47%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2019 - 2023)

- Designed and maintained PCB Debugging data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using MQTT saving \$10k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [Python, PCB Debugging, MQTT, ARM Cortex]
Engineered end-to-end production solution leveraging Python and PCB Debugging with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [MQTT, ARM Cortex, Embedded Linux, I2C/SPI/UART]
Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Computer Science

TUM Munich — 2015 - 2019 (Score: 3.76 / 4.0)